Two-dimensional self-avoiding walks

Bachelor's Thesis in Physics – University of Pisa

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This thesis investigates planar self-avoiding walks as a simplified model of polymer conformations in a good solvent. The polymer is represented by a path on a lattice, with each visited site corresponding to a monomer. Prohibiting repeated visits to the same site incorporates excluded volume, since distinct parts of the chain cannot occupy the same position. Despite omitting molecular detail, the model captures the collective behaviour responsible for the swollen conformations of long polymer chains.

Two sampling strategies are examined. The first, direct rejection, generates independent random walks step by step, discarding each path as soon as it revisits an occupied site. At a fixed chain length, the surviving walks form an unbiased sample of self-avoiding configurations. Their survival probability, however, decreases rapidly as the chain becomes longer, limiting the range of accessible lengths. The method nevertheless provides a simple introduction to the model and to the limitations of direct generation.

The second strategy, the Pivot algorithm, operates on an existing self-avoiding configuration. An internal site is selected as a pivot, and the subsequent portion of the chain undergoes a lattice symmetry transformation, such as a rotation or reflection. The proposal is accepted only if self-avoidance is preserved; otherwise, the previous configuration is retained. Repeated updates generate a Markov chain, from which measurements are collected after an initial thermalization stage. Since the Pivot algorithm modifies an already valid configuration rather than reconstructing a walk from the beginning, its structure allows substantially longer chains to be sampled.

The main physical observable is the mean squared end-to-end distance, which measures the spatial extension of the polymer as a function of chain length. The aim is to determine this quantity over the range accessible to each method and verify whether the numerical results reproduce the expected theoretical behaviour. For sufficiently long self-avoiding walks in two dimensions, the mean squared end-to-end distance is expected to follow an asymptotic power-law dependence on chain length.

Failure probability is also considered as a diagnostic. In direct rejection, failure occurs when the next step reaches an already occupied site, causing the configuration to be discarded. In the Pivot algorithm, failure occurs when a proposed transformation creates an intersection, in which case the current configuration is retained.

A natural extension would be to investigate self-avoiding walks in three dimensions, thereby bringing the model closer to the geometry of real polymer systems. A further development would be to introduce interactions between nonconsecutive monomers occupying neighbouring lattice sites, thus defining an interacting self-avoiding walk. This would make it possible to account for variations in solvent quality and to explore the transition between swollen and compact polymer conformations.

Bibliography

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